MATH 1210 Worksheet 7 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

`http://home.cc.umanitoba.ca/~thomas/Courses`

1. (a) Is an identity matrix invertible? If yes, what is its inverse?
   (b) Is a square zero matrix invertible? If yes, what is its inverse?
   (c) Suppose that A is a square matrix such that $A^n = 0$ for some integer $n \geq 2$. Show that A is not invertible.
2. For a real number t consider the matrix
$$A = \begin{pmatrix} t & 0 & 6 \\ 1 & 3 & 4 \\ 0 & 1 & -1 \end{pmatrix}.$$
   (a) Find the value(s) of t for which A is invertible.
   (b) Use the adjoint method to compute A^{-1} for $t = 1$.
   (c) Use the direct method to compute A^{-1} for $t = 1$.
3. Consider the matrix
$$A = \begin{pmatrix} 2 & 1 & -1 \\ 5 & 0 & 3 \\ t & 2 & 2 \end{pmatrix}$$
   where t is a real number. It is given to you that A is invertible and that the entry in the second column and third row of A^{-1} is $1/7$. Use the adjoint method to find t.
4. (a) Use any method to compute the inverse of
$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 2 & 1 \\ 5 & 1 & -1 \end{pmatrix}.$$
   (b) Use the result of part (a) to solve the system
$$\begin{aligned} 2x + 5z &= 2 \\ 3x + 2y - z &= -1 \\ 5x + y - z &= -1 \end{aligned}$$
   (c) Are the vectors $\mathbf{u}_1 = \langle 2, 0, 5\rangle$, $\mathbf{u}_2 = \langle 3, 2, 1\rangle$ and $\mathbf{u}_3 = \langle 1, 1, -1\rangle$ linearly dependent?
5. Consider the vectors $\mathbf{u_1} = \langle 1, 2, 3, 6\rangle$, $\mathbf{u_2} = \langle 5, -2, 5, -4\rangle$, $\mathbf{u_3} = \langle 3, 3, 2, 1\rangle$ and
$$\mathbf{u_4} = 2\mathbf{u_1} - 3\mathbf{u_2} + \mathbf{u_3} = \langle -10, 13, -7, 25\rangle.$$
   Without doing any computation, is the matrix
$$A = \begin{pmatrix} 1 & 2 & 3 & 6 \\ 5 & -2 & 5 & -4 \\ 3 & 3 & 2 & 1 \\ -10 & 13 & -7 & 25 \end{pmatrix}$$
   invertible?

6. Consider the linear map $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ sending $\langle 1, 1\rangle$ to $\langle -1, 4\rangle$, and $\langle 0, 1\rangle$ to $\langle -2, 3\rangle$.
   (a) Use linearity to find the image of $\langle 1, 0\rangle$.
   (b) Find the associated matrix.
   (c) Find the eigenvalues and eigenvectors of T.
7. Consider the linear map $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ given by
$$\begin{aligned} v_1' &= v_1 + 2v_2 - v_3 \\ v_2' &= 5v_1 + 4v_2 + v_3 \\ v_3' &= 2v_1 + v_2. \end{aligned}$$
   (a) Compute $T(\mathbf{v})$ for $\mathbf{v} = \langle 1, 2, -1\rangle$.
   (b) Find $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ such that $T(\mathbf{v}) = \langle 3, 3, -1\rangle$.
8. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be the linear transformation such that $T(2\mathbf{i}) = \langle 2, -4, 6\rangle$, $T(-\mathbf{j}) = \langle 3, 1, -2\rangle$ and $T(\mathbf{v}) = \langle 1, 0, 5\rangle$ for $\mathbf{v} = \langle 1, -1, 1\rangle$.
   (a) Use the linearity of T to find $T(\mathbf{k})$.
   (b) Find the matrix A associated with T.
9. Consider the linear transformation $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ that reflects every vector in the plane $x = y$.
   (a) Find the associated matrix.
   (b) Find the image of the vector $\mathbf{v} = \langle 1, 2, 3\rangle$ under T.
   (c) Find $\mathbf{v}$ such that $T(\mathbf{v}) = \langle 2, 0, 1\rangle$.
10. Consider the map that sends a nonzero vector $\mathbf{v}$ to the unit vector in the opposite direction to $\mathbf{v}$. Determine if the map is linear.
11. Consider the matrix
$$A = \begin{pmatrix} -1 & 0 & t \\ 3 & 3 & 3 \\ 2 & 0 & 5 \end{pmatrix}$$
    where t is a real number.
    (a) Without doing any computation, explain why $\lambda = 3$ is always eigenvalue, for any value of t. Find all the corresponding eigenvectors.
    (b) Find the other eigenvalues when $t = 2$.
    (c) For which value(s) of t is $\langle 1, 3, 2\rangle$ an eigenvector to the eigenvalue $\lambda = 6$?
12. Find all eigenvalues of the matrix $A = \begin{pmatrix} 3 & 5 & -5 \\ -2 & 3 & 2 \\ -2 & 5 & 0 \end{pmatrix}$. Find the eigenvectors to the largest eigenvalue.